



Fraud Detection Insurance Claim project

Project Epics

Aa Property	Project Tasks	Q Progress
 <u>Data Acquisition & Understanding</u>		
 <u>Data Cleaning & Preparation</u>	 <u>Handle Missing Values</u>	1
 <u>Feature Engineering & Refined Dataset Creation</u>	 <u>Claim Deviation</u>	1
 <u>Feature Selection & Data Governance</u>	 <u>Separate columns (target col & tested col) feature scaling</u>	1
 <u>EDA (Exploratory Data Analysis).</u>	 <u>Fraud vs non-fraud distribution</u>	1
 <u>Model Development</u>	 <u>Random Forest/Logistic Regression</u>	1
 <u>Model Evaluation</u>	 <u>Accuracy</u>	1

Aa Property	Project Tasks	Progress
 <u>Prediction & Insights Generation</u>	 <u>Fraud probability score</u>	1
 <u>Dashboard & Visualization</u>	 <u>Fraud trends</u>	1
 <u>Deployment / Business Integration (Optional Advanced Epic)</u>	 <u>Export model results</u>	1

Project Tasks

Project Epics	Aa Tasks	checkbox
 <u>Data Cleaning & Preparation</u>	 <u>Handle Missing Values</u>	☒
	 <u>Remove duplicates</u>	☒
	 <u>Standardize Format</u>	☒
	 <u>Visualizing Missing Values</u>	☒
	 <u>Encode categorical variables</u>	☒
 <u>Feature Engineering & Refined Dataset Creation</u>	 <u>Claim Deviation</u>	☒
	 <u>Average claim amount</u>	☐
	 <u>Claim Ratio</u>	☒
	 <u>Derived flags (high claim, abnormal patterns)</u>	☐
 <u>Feature Selection & Data Governance</u>	 <u>Separate columns (target col & tested col) feature scaling</u>	☒
	 <u>Remove unnecessary columns (identifiers)</u>	☒
	 <u>Label Encoding & One</u>	☒

Project Epics	Tasks	checkbox
	<u>hot encoding (if req</u>	
	<u>Eliminate leakage columns (e.g., total_claim_amount)</u>	✓
	<u>Multicollinearity checking</u>	✓
<u>EDA (Exploratory Data Analysis)</u>	<u>Fraud vs non-fraud distribution</u>	✓
	<u>Claim amount analysis</u>	✓
	<u>Outliers Detection</u>	
	<u>Correlation analysis</u>	✓
<u>Model Development</u>	<u>Random Forest/Logistic Regression</u>	✓
	<u>Train-test split</u>	✓
	<u>Hyperparameter tuning (optional advanced)</u>	
<u>Model Evaluation</u>	<u>Accuracy</u>	✓
	<u>precision</u>	✓
	<u>confusion matrix</u>	✓
	<u>classification Report</u>	✓
	<u>Recall (priority)</u>	✓
<u>Prediction & Insights Generation</u>	<u>Fraud probability score</u>	✓
	<u>Fraud classification</u>	✓
	<u>Feature importance analysis</u>	✓
<u>Dashboard & Visualization</u>	<u>Fraud trends</u>	✓
	<u>High-risk providers</u>	✓
	<u>Claim distribution</u>	✓
	<u>Top Fraud Drivers</u>	✓

↗ Project Epics	Aa Tasks	checkbox
	<u>Classification Results: Fraud vs Non-Fraud</u>	☒
	<u>Decision tree Model</u>	☒
	<u>Random Tree Model</u>	☒
	 <u>Filters (time, provider, patient)</u>	☐
 <u>Deployment / Business Integration (Optional Advanced Epic)</u>	 <u>Export model results</u>	☒
	 <u>Integrate with reporting systems</u>	☒
	 <u>Set up periodic model refresh</u>	☒
	<u>Journal - Thursday, April 16, 2026</u>	☐
	<u>Journal - Friday, April 17, 2026</u>	☐

☐ Pending Tasks

Calendar

Aa Name
<u>Untitled</u>